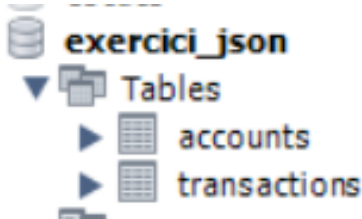


<https://github.com/AndrewUser950/DAW-LMSGI-AndreuCompany/tree/main/Tasca%20Avaluable%207.4>

1. Carrega la informació dels arxius Transactions.json i Accounts.json al mysql workbench en dues taules. Explica tot el procés amb captures de pantalla i copia el codi.

Per començar, he creat ses taules dels arxius a dues taules diferents dins una base de dades creada per a s'exercici.

```
1 • CREATE DATABASE IF NOT EXISTS exercici_json;
2 • USE exercici_json;
3
4 • CREATE TABLE accounts (
5     id INT AUTO_INCREMENT PRIMARY KEY,
6     data JSON
7 );
8
9 • CREATE TABLE transactions (
10    id INT AUTO_INCREMENT PRIMARY KEY,
11    data JSON
12 );
```



Ara que ja tenc ses taules, hem de passar a inserir es contingut dels arxius descarregats per poder fer ses consultes.

Per inserir ses dades, he decidit anar copiant i aferrant tota sa informació per després inserir-la.



```
1 • INSERT INTO transactions (data) VALUES ('{"txn_id": "1", "amount": 100, "funds_avail_date": "2000-01-15 00:00:00", "txn_c
2 • INSERT INTO transactions (data) VALUES ('{"txn_id": "2", "amount": 500, "funds_avail_date": "2000-01-15 00:00:00", "txn_c
3 • INSERT INTO transactions (data) VALUES ('{"txn_id": "3", "amount": 75, "funds_avail_date": "2004-06-30 00:00:00", "txn_di
4 • INSERT INTO transactions (data) VALUES ('{"txn_id": "4", "amount": 1125, "funds_avail_date": "2001-03-12 00:00:00", "txn
5 • INSERT INTO transactions (data) VALUES ('{"txn_id": "5", "amount": 250, "funds_avail_date": "2001-03-12 00:00:00", "txn_c
6 • INSERT INTO transactions (data) VALUES ('{"txn_id": "6", "amount": 325, "funds_avail_date": "2002-11-23 00:00:00", "txn_c
7 • INSERT INTO transactions (data) VALUES ('{"txn_id": "7", "amount": 350, "funds_avail_date": "2002-12-15 00:00:00", "txn_c
8 • INSERT INTO transactions (data) VALUES ('{"txn_id": "8", "amount": 154, "funds_avail_date": "2003-09-12 00:00:00", "txn_c
9 • INSERT INTO transactions (data) VALUES ('{"txn_id": "9", "amount": 55, "funds_avail_date": "2000-01-15 00:00:00", "txn_di
10 • INSERT INTO transactions (data) VALUES ('{"txn_id": "10", "amount": 452, "funds_avail_date": "2004-09-30 00:00:00", "txn
11 • SELECT * FROM transactions;
```

Quan ho he tengut tot inserit i llest, li he donat a s'icona des raig per poder executar totes ses línies i inserir sa informació dins ses taules.

CODI:

```
CREATE DATABASE IF NOT EXISTS exercici_json;  
USE exercici_json;
```

```
CREATE TABLE accounts (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    data JSON  
);
```

```
CREATE TABLE transactions (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    data JSON  
);
```

Insert accounts

```
INSERT INTO accounts (data) VALUES ('{"account_id": "1", "avail_balance": 1057.75,  
"close_date": "NULL", "last_activity_date": "2005-01-04", "open_date": "2000-01-15",  
"pending_balance": 1057.75, "status": "ACTIVE", "cust_id": 1, "open_emp_id": 10,  
"open_branch_id": "2", "open_branch_name": "Woburn Branch", "open_branch_address":  
"422 Maple St.", "open_branch_city": "Woburn", "open_branch_state": "MA",  
"open_branch_zip_code": "01801", "product_cd": "CHK", "product_date_offered":  
"2000-01-01", "product_date_retired": "NULL", "product_name": "checking account",  
"product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}');  
INSERT INTO accounts (data) VALUES ('{"account_id": "2", "avail_balance": 500,  
"close_date": "NULL", "last_activity_date": "2004-12-19", "open_date": "2000-01-15",  
"pending_balance": 500, "status": "ACTIVE", "cust_id": 1, "open_emp_id": 10,  
"open_branch_id": "2", "open_branch_name": "Woburn Branch", "open_branch_address":  
"422 Maple St.", "open_branch_city": "Woburn", "open_branch_state": "MA",  
"open_branch_zip_code": "01801", "product_cd": "SAV", "product_date_offered":  
"2000-01-01", "product_date_retired": "NULL", "product_name": "savings account",  
"product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}');  
INSERT INTO accounts (data) VALUES ('{"account_id": "3", "avail_balance": 3000,  
"close_date": "NULL", "last_activity_date": "2004-06-30", "open_date": "2004-06-30",  
"pending_balance": 3000, "status": "ACTIVE", "cust_id": 1, "open_emp_id": 10,  
"open_branch_id": "2", "open_branch_name": "Woburn Branch", "open_branch_address":  
"422 Maple St.", "open_branch_city": "Woburn", "open_branch_state": "MA",  
"open_branch_zip_code": "01801", "product_cd": "CD", "product_date_offered":  
"2000-01-01", "product_date_retired": "NULL", "product_name": "certificate of deposit",  
"product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}');  
INSERT INTO accounts (data) VALUES ('{"account_id": "4", "avail_balance": 2258.02,  
"close_date": "NULL", "last_activity_date": "2004-12-27", "open_date": "2001-03-12",  
"pending_balance": 2258.02, "status": "ACTIVE", "cust_id": 2, "open_emp_id": 10,  
"open_branch_id": "2", "open_branch_name": "Woburn Branch", "open_branch_address":  
"422 Maple St.", "open_branch_city": "Woburn", "open_branch_state": "MA",
```

```
"open_branch_zip_code": "01801", "product_cd": "CHK", "product_date_offered":  
"2000-01-01", "product_date_retired": "NULL", "product_name": "checking account",  
"product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}');  
INSERT INTO accounts (data) VALUES ('{"account_id": "5", "avail_balance": 200,  
"close_date": "NULL", "last_activity_date": "2004-12-11", "open_date": "2001-03-12",  
"pending_balance": 200, "status": "ACTIVE", "cust_id": 2, "open_emp_id": 10,  
"open_branch_id": "2", "open_branch_name": "Woburn Branch", "open_branch_address":  
"422 Maple St.", "open_branch_city": "Woburn", "open_branch_state": "MA",  
"open_branch_zip_code": "01801", "product_cd": "SAV", "product_date_offered":  
"2000-01-01", "product_date_retired": "NULL", "product_name": "savings account",  
"product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}');  
INSERT INTO accounts (data) VALUES ('{"account_id": "7", "avail_balance": 1057.75,  
"close_date": "NULL", "last_activity_date": "2004-11-30", "open_date": "2002-11-23",  
"pending_balance": 1057.75, "status": "ACTIVE", "cust_id": 3, "open_emp_id": 13,  
"open_branch_id": "3", "open_branch_name": "Quincy Branch", "open_branch_address":  
"125 Presidential Way", "open_branch_city": "Quincy", "open_branch_state": "MA",  
"open_branch_zip_code": "02169", "product_cd": "CHK", "product_date_offered":  
"2000-01-01", "product_date_retired": "NULL", "product_name": "checking account",  
"product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}');  
SELECT * FROM accounts;
```

Insert transactions

```
INSERT INTO transactions (data) VALUES ('{"txn_id": "1", "amount": 100,  
"funds_avail_date": "2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00",  
"txn_type_cd": "CDT", "account_id": 1, "teller_emp_id": 1, "execution_branch_id": 1,  
"execution_branch_name": "Headquarters", "execution_branch_address": "3882 Main St.",  
"execution_branch_city": "Waltham", "execution_branch_state": "MA",  
"execution_branch_zip_code": "02451"}');  
INSERT INTO transactions (data) VALUES ('{"txn_id": "2", "amount": 500,  
"funds_avail_date": "2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00",  
"txn_type_cd": "CDT", "account_id": 2, "teller_emp_id": 2, "execution_branch_id": 1,  
"execution_branch_name": "Headquarters", "execution_branch_address": "3882 Main St.",  
"execution_branch_city": "Waltham", "execution_branch_state": "MA",  
"execution_branch_zip_code": "02451"}');  
INSERT INTO transactions (data) VALUES ('{"txn_id": "3", "amount": 75, "funds_avail_date":  
"2004-06-30 00:00:00", "txn_date": "2004-06-30 00:00:00", "txn_type_cd": "CDT",  
"account_id": 3, "teller_emp_id": 2, "execution_branch_id": 2, "execution_branch_name":  
"Woburn Branch", "execution_branch_address": "422 Maple St.", "execution_branch_city":  
"Woburn", "execution_branch_state": "MA", "execution_branch_zip_code": "01801"}');  
INSERT INTO transactions (data) VALUES ('{"txn_id": "4", "amount": 1125,  
"funds_avail_date": "2001-03-12 00:00:00", "txn_date": "2001-03-12 00:00:00",  
"txn_type_cd": "CDT", "account_id": 4, "teller_emp_id": 2, "execution_branch_id": 1,  
"execution_branch_name": "Headquarters", "execution_branch_address": "3882 Main St.",
```

```
"execution_branch_city": "Waltham", "execution_branch_state": "MA",
"execution_branch_zip_code": "02451"});
INSERT INTO transactions (data) VALUES ('{"txn_id": "5", "amount": 250,
"funds_avail_date": "2001-03-12 00:00:00", "txn_date": "2001-03-12 00:00:00",
"txn_type_cd": "CDT", "account_id": 5, "teller_emp_id": 7, "execution_branch_id": null,
"execution_branch_name": null, "execution_branch_address": null, "execution_branch_city":
null, "execution_branch_state": null, "execution_branch_zip_code": null}');
INSERT INTO transactions (data) VALUES ('{"txn_id": "6", "amount": 325,
"funds_avail_date": "2002-11-23 00:00:00", "txn_date": "2002-11-23 00:00:00",
"txn_type_cd": "CDT", "account_id": 7, "teller_emp_id": 9, "execution_branch_id": null,
"execution_branch_name": null, "execution_branch_address": null, "execution_branch_city":
null, "execution_branch_state": null, "execution_branch_zip_code": null}');
INSERT INTO transactions (data) VALUES ('{"txn_id": "7", "amount": 350,
"funds_avail_date": "2002-12-15 00:00:00", "txn_date": "2002-12-15 00:00:00",
"txn_type_cd": "CDT", "account_id": 8, "teller_emp_id": 7, "execution_branch_id": 4,
"execution_branch_name": "So. NH Branch", "execution_branch_address": "378 Maynard
Ln.", "execution_branch_city": "Salem", "execution_branch_state": "NH",
"execution_branch_zip_code": "03079"}');
INSERT INTO transactions (data) VALUES ('{"txn_id": "8", "amount": 154,
"funds_avail_date": "2003-09-12 00:00:00", "txn_date": "2003-09-12 00:00:00",
"txn_type_cd": "CDT", "account_id": 10, "teller_emp_id": 12, "execution_branch_id": null,
"execution_branch_name": null, "execution_branch_address": null, "execution_branch_city":
null, "execution_branch_state": null, "execution_branch_zip_code": null}');
INSERT INTO transactions (data) VALUES ('{"txn_id": "9", "amount": 55, "funds_avail_date":
"2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00", "txn_type_cd": "CDT",
"account_id": 11, "teller_emp_id": 9, "execution_branch_id": 4, "execution_branch_name":
"So. NH Branch", "execution_branch_address": "378 Maynard Ln.", "execution_branch_city":
"Salem", "execution_branch_state": "NH", "execution_branch_zip_code": "03079"}');
INSERT INTO transactions (data) VALUES ('{"txn_id": "10", "amount": 452,
"funds_avail_date": "2004-09-30 00:00:00", "txn_date": "2004-09-30 00:00:00",
"txn_type_cd": "CDT", "account_id": 12, "teller_emp_id": 13, "execution_branch_id": 2,
"execution_branch_name": "Woburn Branch", "execution_branch_address": "422 Maple St.",
"execution_branch_city": "Woburn", "execution_branch_state": "MA",
"execution_branch_zip_code": "01801"}');
SELECT * FROM transactions;
```

2. Troba la consulta SQL utilitzant les funcions proporcionades per MySQL per a manegar codi JSON que retornaria la solució a les següents preguntes.

2.1. Quina consulta utilitzaries per obtenir l'import (amount) de totes les transaccions de la taula transactions (Utilitza la funció JSON_UNQUOTE per a eliminar les cometes)?

```
SELECT JSON_UNQUOTE(JSON_EXTRACT(data, '$.amount')) AS amount  
FROM transactions;
```

2.2. Com podries filtrar les transaccions que tenen una quantitat superior a 200 (utilitza JSON_EXTRACT)?

```
SELECT *  
FROM transactions  
WHERE JSON_EXTRACT(data, '$.amount') > 200;
```

2.3. Escriu una consulta que obtingui l'ID de la transacció (txn_id) i la data de la transacció (txn_date) per totes les transaccions.

```
SELECT  
    JSON_EXTRACT(data, '$.txn_id') AS txn_id,  
    JSON_EXTRACT(data, '$.txn_date') AS txn_date  
FROM transactions;
```

2.4. Escriu una consulta que recuperi l'ID del compte (account_id) i l'import (amount) de les transaccions, unint les taules transactions i accounts per account_id.

```
SELECT  
    JSON_EXTRACT(t.data, '$.account_id') AS account_id,  
    JSON_EXTRACT(t.data, '$.amount') AS amount  
FROM transactions t  
JOIN accounts a  
    ON JSON_UNQUOTE(JSON_EXTRACT(t.data, '$.account_id')) =  
    JSON_UNQUOTE(JSON_EXTRACT(a.data, '$.account_id'));
```

2.5. Com podries calcular l'import total de les transaccions per cada compte (account_id)?

```
SELECT  
    JSON_EXTRACT(data, '$.account_id') AS account_id,  
    SUM(CAST(JSON_EXTRACT(data, '$.amount') AS DECIMAL(10,2))) AS total_import  
FROM transactions  
GROUP BY JSON_EXTRACT(data, '$.account_id');
```